

Session Objectives

- **Identify sedative-hypnotic drugs.**
- **Describe MOA, use, and adverse effects of barbiturates.**
- **Describe MOA and adverse effects, and specific uses of benzodiazepines (BDZ).**
- **Describe the differences between barbs, BDZs, and ethanol in MOA and adverse effects. Include management of overdose, withdrawal, and abstinence.**
- **Describe non-sedating anxiolytic drugs.**
- **Describe drugs used to induce sleep.**
- **Describe drugs for narcolepsy.**

Sedative-Hypnotic Drugs

- Barbiturates
- Benzodiazepines
- Ethanol
- Non-benzodiazepines (zolpidem, zaleplon, eszopiclone)
- 1st generation antihistamines
- Newer drugs (melatonin agonists, orexin antagonists)

Degree of sedation-hypnosis depends on dose

Barbiturates

- Pentobarbital, Secobarbital, Phenobarbital, Amobarbital, Thiopental
- Lipid soluble drugs (thiopental) are **redistributed** from brain.
- Elderly and those with hepatic disease have increase in $t_{1/2}$.
- **CYP inducers (3A4)**
- Exacerbation of porphyria
- Use: anxiety and insomnia (fallen out of use), seizures (phenobarbital), induction of anesthesia (thiopental).

Barbiturates

- Increase **duration** of GABA action. Opens the chloride gate in the **absence** of GABA.
- “No ceiling effect”
- Induce full respiratory arrest.
- Provide supportive care alkalize urine

Benzodiazepines

- Chlordiazepoxide and Diazepam (long acting). Alcohol detoxification.
- Alprazolam (short acting). Panic disorder.
- Lorazepam, Clonazepam, Diazepam (seizure disorder). **1st line for Status epilepticus**
- Midazolam (anesthesia)
- Remimazolam (newest; rapid onset and offset)
- Estazolam and Flurazepam (insomnia)

Benzodiazepines

- **Liver metabolism (converted to active metabolites).**
- **Chlordiazepoxide, diazepam, flurazepam (metabolites make the drug long acting). Appropriate for detoxification protocols.**
- **Oxazepam, temazepam, and lorazepam are safest for elderly. Bypass phase I metabolism (only conjugated).**
- **Use all benzos cautiously in elderly.**

Benzodiazepines

- Potentiates GABA by increasing **frequency** of chloride channel opening. Needs GABA to work.
- “Ceiling effect”
- No respiratory depression, unless combined with **other** CNS depressants.
- Use: anxiety, sedative-hypnotic, insomnia, anterograde amnesia, muscle relaxant, anticonvulsant, anticipatory N/V

Benzodiazepines Adverse Effects

- **Drowsiness, hangover**
- **Mild euphoria (abuse potential)**
- **Anterograde amnesia**
- **Physical dependence: tolerance and withdrawal symptoms: rebound anxiety, insomnia, seizures (alprazolam).**
- **Fetal malformations “floppy infant”**

Flumazenil

- **Benzodiazepine antagonist, IV**
- **Reverses benzodiazepine toxicity in overdose.**
- **When flumazenil is used after midazolam, this drug will rapidly result in patient recovery.**
- **Caution in chronic BDZ users, precipitates “withdrawal”**

Non-Benzodiazepines “z drugs”

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- Zolpidem, zaleplon, eszopiclone
- Minimal effect on REM sleep (unlike barbs and BDZ).
- Shorter duration of action
- Less hangover effect
- Target specific GABA_A subunit.
- Potentiated by ethanol

Other “Sedative” Drugs

- Suvorexant, lemborexant, and daridorexant (orexin antagonists at OX1, OX2 receptors)
- **H₁ receptor antagonists (1st Generation only)**
Adverse effects?
- Tricyclic antidepressants (doxepin)
- Trazodone (H₁ blockade)
- Ethanol (avoid with other CNS “depressants”)

Ethanol (chronic)

- Dilated cardiomyopathy
- Thiamine deficiency (Wernicke's encephalopathy): thiamine before glucose.
- Fetal alcohol syndrome
- Zero order kinetics
- Induces CYP 2E1 (enhanced acetaminophen toxicity)
- Increased inflammation/neoplasia
- Loss of liver function

Ethanol Intoxication

- Slurred speech, nystagmus, stupor, or coma.
- Hypotension and tachycardia
- Hypoglycemia, lactic acidosis, hypokalemia, hypomagnesemia
- Dehydration
- Treatment: supportive care, ventilation
- 80mg/dL BAC (intoxication)

Ethanol Overdose, Withdrawal, and Alcohol Use Disorder

- Overdose: maintain airway, supportive care
- Symptoms of withdrawal: CNS stimulation, seizures, “delirium tremens” BENZO must be administered. Long-acting Benzo (ex. Chlordiazepoxide).
- Alcohol use disorder: Naltrexone (opioid blocker); Acamprosate (restores GABA/glutamate balance); CBT

Anxiety Classification and Treatment

- **Acute anxiety: Benzodiazepine**
- **Panic disorder: Alprazolam, then SSRI for long-term.**
- **Phobic disorder: Benzo then SSRI.**
- **Propranolol (“Situational anxiety”)**
- **OCD: SSRI and psychotherapy**
- **Generalized anxiety disorder (GAD): Benzo, then SSRI, SNRI, or buspirone**

Buspirone

- Partial 5HT_{1A} agonist, not GABA.
- Weak D₂ antagonist
- Use: chronic anxiety (GAD)
- Reduces anxiety in absence of tolerance, rebound, or dependence.
- Full anxiolytic effect may be 4-6 weeks
- Adverse effects: dizziness

Narcolepsy

- Daytime sleepiness, despite sleeping at night
- Treat with sympathomimetics “stimulants”, also for ADHD
- Amphetamine, dextroamphetamine, and methylphenidate inhibit reuptake of NE and DA. Also increases release of catecholamines from nerve terminals.
- Adverse effects: Cardiovascular, weight loss, anxiety.
- Modafinil and armodafinil (greater DA vs NE reuptake inhibition and release). Less anxiety and anorexia than amphetamines. Skin rash reported. **Activates orexin, inhibits GABA.**

Stimulant Overdose

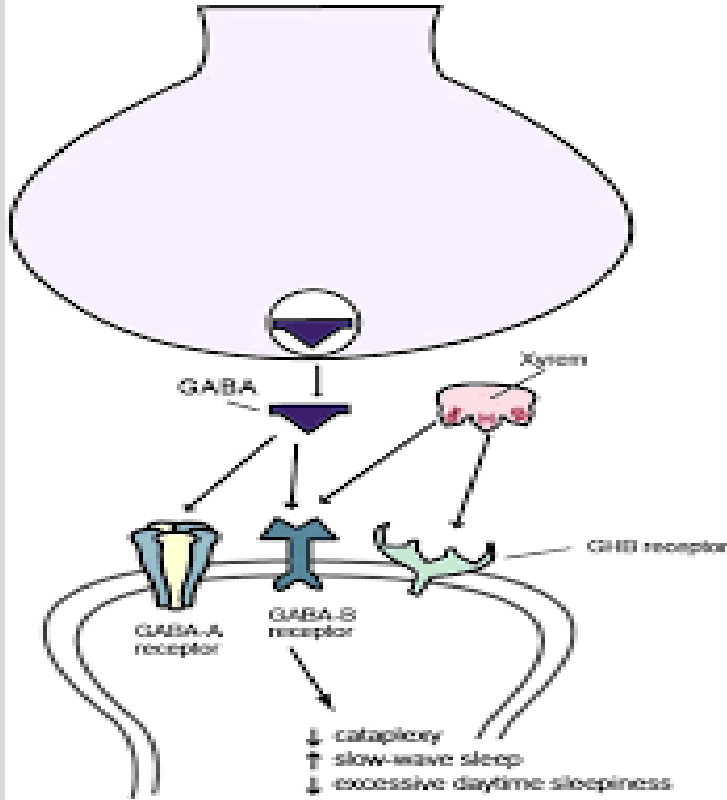
- **Symptoms of overdose: difficult to differentiate schizophrenia symptoms, tachycardia, mydriasis, hypertension, hyperthermia, seizures.**
- **“Cocaine bugs”**
- **Overdose: acidify urine, supportive therapy (lorazepam, haloperidol, and vasodilator). Labetalol**

Sodium Oxybate

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Mechanism of Action of
Sodium Oxybate (Xyrem, GHB)



- Treats narcolepsy with cataplexy.
- Sodium salt of gamma-hydroxybutyrate (GHB), metabolite of GABA.
- Agonist of GABA_B receptor, short $t_{1/2}$
- **Risk Evaluation and Mitigation (REMS) program.** Mitigates the risk of abuse and misuse.
- AE: N/V, confusion, not to be administered with ethanol or sedative hypnotics (increased risk of respiratory depression).
- Stimulant may be taken during day